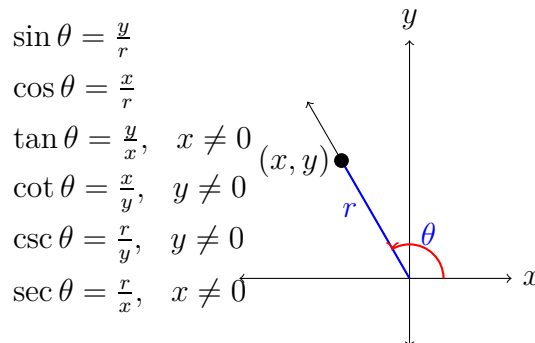


DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I

WEEK 11: LECTURE NOTE

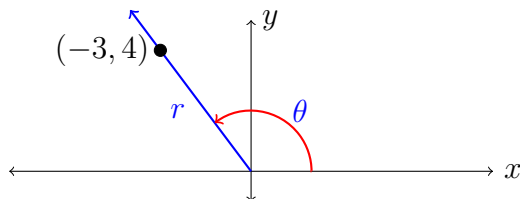
TRIGONOMETRIC FUNCTIONS OF ANY ANGLE

Definition 1 (Trigonometric Functions of Any Angle). Let θ be an angle in standard position with a point (x, y) on the terminal side of θ and $r = \sqrt{x^2 + y^2} \neq 0$.



Because $r \neq 0$, it follows that the **sine** and **cosine** functions are defined for any real value of θ .

Example 2. Let $(-3, 4)$ be a point on the terminal side of θ as shown on the figure below. Find the sine, cosine, and tangent of θ



Solution. Referring to the Figure above, we can see that $x = -3$, $y = 4$ and

$$r = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

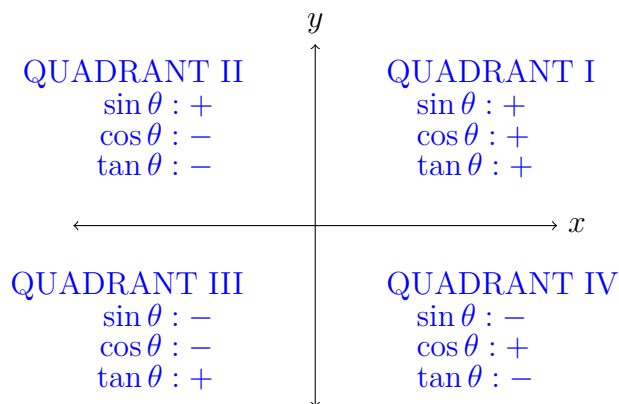
Therefore,

$$\sin \theta = \frac{y}{r} = \frac{4}{5} = 0.8$$

$$\cos \theta = \frac{x}{r} = \frac{-3}{5} = -0.6$$

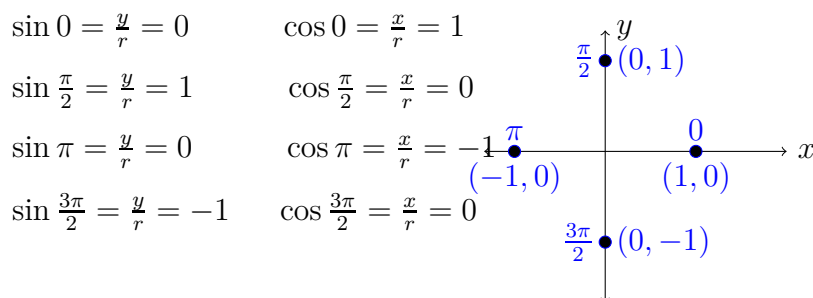
$$\tan \theta = \frac{y}{x} = \frac{4}{-3} = -\frac{4}{3} \approx -1.333$$

The signs of the trigonometric functions in the four quadrants can be determined easily from the definitions of the functions. Since $r > 0$, we note that $\sin \theta$ is positive wherever $y > 0$ and $\cos \theta$ is positive wherever $x > 0$. The signs of other four (if defined at θ) trigonometric functions are determined in terms of signs of $\sin \theta$ and $\cos \theta$.



Definition 3 (Trigonometric Functions of Quadrant Angles). An angle θ in standard position is said to be a **quadrant angle** if its terminal side lies on one of the coordinate axes. These angles are integral multiples of the angles $0^\circ = 0 \text{ rad}$, $90^\circ = \frac{\pi}{2} \text{ rad}$, $180^\circ = \pi \text{ rad}$, and $270^\circ = \frac{3\pi}{2} \text{ rad}$.

Consider the points $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$ on the terminal side of the quadrant angles 0 , $\frac{\pi}{2}$, π and $\frac{3\pi}{2}$, respectively. Then



Example 4. Given $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, find $\cos \theta$, $\tan \theta$, $\csc \theta$, and $\sec \theta$.

Solution. Note that θ lies in Quadrant III. Since $\sin \theta = -\frac{2}{3} = \frac{y}{r}$ and $y < 0$, we have $y = -2$ and $r = 3$. Hence $x^2 + y^2 = r^2 \Rightarrow x^2 + 4 = 9 \Rightarrow x^2 = 5 \Rightarrow x = \pm\sqrt{5}$.

Since (x, y) is in Quadrant III, $x < 0$; so that $x = -\sqrt{5}$. Therefore,

$$(1) \cos \theta = \frac{x}{r} = -\frac{\sqrt{5}}{3} \quad (2) \tan \theta = \frac{y}{x} = \frac{-2}{-\sqrt{5}} = \frac{2\sqrt{5}}{5}$$

$$(2) \csc \theta = \frac{r}{y} = -\frac{3}{2} \quad (3) \sec \theta = \frac{r}{x} = -\frac{3}{\sqrt{5}} = -\frac{3\sqrt{5}}{5}$$

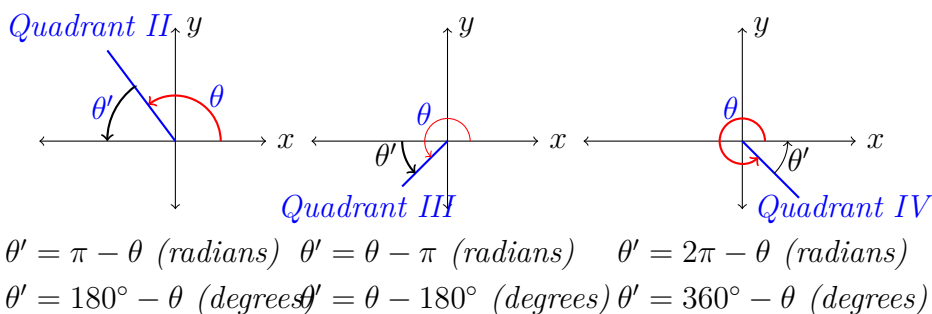
Trigonometric Values of Common Angles: $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ, 180^\circ, 270^\circ$.

θ in degrees	0°	30°	45°	60°	90°	180°	270°
θ in radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1
$\cos \theta$	1	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	0	-1	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	1	Undefined	0	Undefined

REFERENCE ANGLES

The values of the trigonometric functions of angles greater than 90° (or less than 0°) can be determined from their values at corresponding acute angles called reference angles.

Definition 5 (REFERENCE ANGLE). Let θ be an angle in standard position. Its **reference angle** θ' is the acute angle formed by the terminal side of θ and the horizontal axis. The reference angles for θ in Quadrants II, III, and IV.



Example 6 (Finding Reference Angles). Find the reference angle θ' . (a) $\theta = 300^\circ$ (b) $\theta = 2.3$ rad (c) $\theta = -135^\circ$

Solution.

(a) Because $\theta = 300^\circ$ lies in Quadrant IV, the angle it makes with the x -axis is

$$\theta' = 360^\circ - 300^\circ = 60^\circ. \quad (\text{Degrees})$$

(b) Because $\frac{\pi}{2} \approx 1.5708 < 2.3 < 3.14 \approx \pi$, $\theta = 2.3$ rad is in Quadrant II and its reference angle is

$$\theta' = \pi - \theta = \pi - 2.3 \approx 0.8416 \quad (\text{Radians})$$

(c) First, determine that $\theta = -135^\circ$ is coterminal with $\theta = 225^\circ$ which lies in Quadrant III. So, the reference angle is

$$\theta' = 225^\circ - 180^\circ = 45^\circ. \quad (\text{Degrees})$$

Definition 7 (Evaluating Trigonometric Functions of Any Angle). To find the value of a trigonometric function of any angle θ :

1. Determine the function value of the associated reference angle θ' .
2. Depending on the quadrant in which θ lies, affix the appropriate sign to the function value of θ' .

Remark 8. The fundamental trigonometric identities listed in the preceding lessons (for an acute angle θ) are also valid when θ is any angle in the domain of the function.

Example 9. Evaluate each trigonometric function.

$$(a) \cos \frac{4\pi}{3} \quad (b) \tan -210^\circ \quad (c) \csc \frac{11\pi}{4} \quad (d) \sin \frac{-20\pi}{3}$$

Solution.

(a) Because $\theta = \frac{4\pi}{3}$ lies in Quadrant III, the reference angle is

$$\theta' = \frac{4\pi}{3} - \pi = \frac{\pi}{3}. \quad (\text{radians})$$

Moreover, the cosine is negative in Quadrant III, so

$$\cos \frac{4\pi}{3} = -\cos \frac{\pi}{3} = -\frac{1}{2}.$$

- (b) Because $360^\circ - 210^\circ = 150^\circ$ it follows that -210° is coterminal with the second-quadrant angle 150° . Therefore, the reference angle is

$$\theta' = 180^\circ - 150^\circ = 30^\circ.$$

Finally, because the tangent is negative in Quadrant II, we have

$$\tan -210^\circ = -\tan 30^\circ = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}.$$

- (c) Because $\frac{11\pi}{4} - 2\pi = \frac{3\pi}{4}$, it follows that $\frac{11\pi}{4}$ is coterminal with the second-quadrant angle $\frac{3\pi}{4}$. Therefore, the reference angle is

$$\theta' = \pi - \frac{3\pi}{4} = \frac{\pi}{4} \quad (\text{radians})$$

Because the cosecant is positive in Quadrant II, we have

$$\csc \frac{11\pi}{4} = \csc \frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

- (d) Because $\frac{-20\pi}{3} + 8\pi = \frac{4\pi}{3}$, it follows that $\frac{-20\pi}{3}$ is coterminal with the third-quadrant angle $\frac{4\pi}{3}$. Therefore, the reference angle is

$$\theta' = \frac{4\pi}{3} - \pi = \frac{\pi}{3} \quad (\text{radians})$$

Because the sine is negative in Quadrant III, we have

$$\sin \frac{-20\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}.$$

Example 10 (Using Trigonometric Identities). *Let θ be an angle in Quadrant II such that $\sin \theta = \frac{1}{3}$. Using trigonometric identities find* (a) $\cos \theta$ (b) $\tan \theta$

Solution.

- (a) Using the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$ we obtain

$$\left(\frac{1}{3}\right)^2 + \cos^2 \theta = 1 \quad \Rightarrow \quad \Rightarrow \quad \cos^2 \theta = 1 - \frac{1}{9} = \frac{8}{9} \quad \Rightarrow \quad \cos \theta = \pm \frac{2\sqrt{2}}{3}.$$

Because $\cos \theta < 0$ in Quadrant II, we can use the negative root to obtain

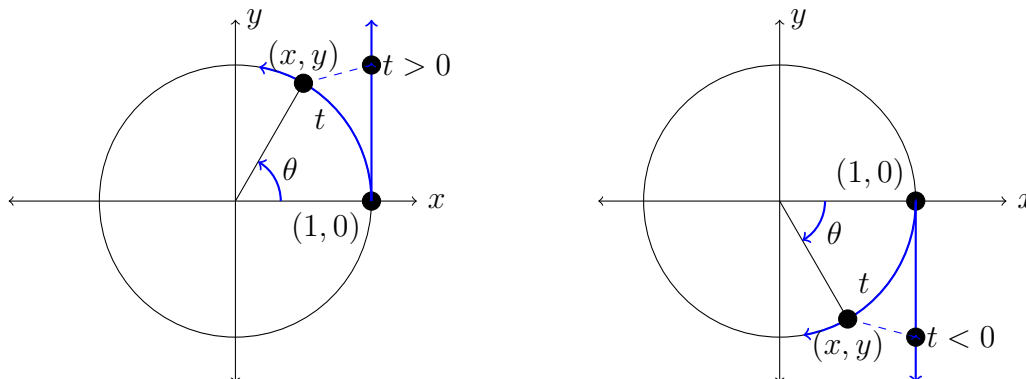
$$\cos \theta = -\frac{2\sqrt{2}}{3}.$$

- (b) Using the trigonometric identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ we obtain

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\frac{1}{3}}{-\frac{2\sqrt{2}}{3}} = -\frac{1}{2\sqrt{2}} = -\frac{\sqrt{2}}{4}.$$

TRIGONOMETRIC FUNCTIONS OF REAL NUMBERS

To define a trigonometric function of a real number (rather than an angle), let t represent any real number. Then imagine that the real number line is wrapped around a unit circle, as shown in the Figure below. Note that positive numbers correspond to a counterclockwise wrapping, and negative numbers correspond to a clockwise wrapping.



As the real number line is wrapped around the unit circle, each real number t will correspond to a central angle θ (in standard position). Moreover, because the circle has a radius of 1, the arc intercepted by the angle θ will have (directional) length $s = r\theta = (1)(t) = t$.

Definition 11 (Trigonometric functions of real numbers). *Let t be a real number. If θ is the central angle in standard position corresponding to t (after wrapping around the unit circle), then we define*

$$\sin t = \sin \theta \quad \text{and} \quad \cos t = \cos \theta.$$

Definition 12 (Periodic Function). *A function f is said to be **periodic** when there exists a positive real number c such that for all t in the domain of f*

$$f(t + c) = f(t)$$

*for all t in the domain of f . The least number for which f is periodic is called the **period of f** .*

If $f(t) = \sin t$ and $g(t) = \cos t$, then $\text{Dom}(f) = (-\infty, +\infty) = \text{Dom}(g)$.

Note that $\theta + 2\pi$ and θ are co-terminal angles for any angle θ in standard position.

Example 13 ($f(x) = \sin x$ is a Periodic Function). *The function $f(x) = \sin x$ is 2π -periodic; that is*

$$\sin(x + 2\pi) = \sin x, \quad x \in \mathbb{R}.$$

Since $(x + 2\pi)\text{rad}$ and $x\text{ rad}$ are co-terminal angles,

$$f(x + 2\pi) = \sin(x + 2\pi) = \sin x = f(x)$$

for all real numbers x .

Example 14 ($f(x) = \cos x$ is a Periodic Function). *The function $f(x) = \cos x$ is 2π -periodic; that is*

$$\cos(x + 2\pi) = \cos x, \quad x \in \mathbb{R}.$$

Remark 15. For any real number x the other four trigonometric functions are defined in terms of the sine and cosine functions. For instance,

1. $\tan x = \frac{\sin x}{\cos x}$ for $x \neq n\pi + \frac{\pi}{2}$ where n is an integer.
2. $\cot x = \frac{\cos x}{\sin x}$ for $x \neq n\pi$ where n is an integer.
3. $\csc x = \frac{1}{\sin x}$ for $x \neq n\pi$ where n is an integer.
4. $\sec x = \frac{1}{\cos x}$ for $x \neq n\pi + \frac{\pi}{2}$ where n is an integer.

Proposition 16 (Even and Odd Trigonometric Functions). *The cosine and secant functions are even. The sine, cosecant, tangent, and cotangent functions are odd.*

1. $\cos(-x) = \cos x$
2. $\sec(-x) = \sec x$
3. $\sin(-x) = -\sin x$
4. $\tan(-x) = -\tan x$
5. $\csc(-x) = -\csc x$
6. $\cot(-x) = -\cot x$

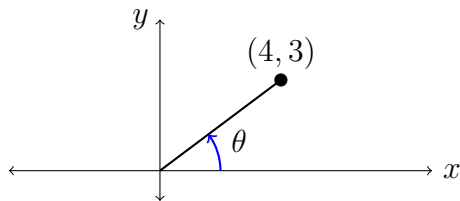
SUMMARY

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- Introduction to basic trigonometry is complete in this stage.
- Measurement of angles in both degrees and radians have been well articulated.
- The definitions of the six trigonometric functions from a right triangle perspective and as functions of real numbers are introduced.

EXERCISES

1. Determine the exact values of the six trigonometric functions of the angle θ shown in the figure below.



2. If $\tan \theta = -\frac{15}{8}$ and $\sin \theta < 0$, find the values of the six trigonometric functions of θ .
3. Evaluate the secant of the quadrant angle $\theta = \pi$ radians, if possible.
4. Find the reference angle θ' of the angle $\theta = -1.8$ (radians). Sketch θ in standard position and label θ' .

- Evaluate the sine, cosine, and tangent of the angle $\theta = -\frac{17\pi}{6}$ without using a calculator.
- Let θ be an angle in Quadrant IV such that $\sin \theta = -\frac{3}{5}$. Find $\cos \theta$.
- Use a calculator to evaluate the trigonometric function. Round your answer to four decimal places. (Be sure the calculator is set in the correct angle mode.)

$$(a) \cot(-220^\circ)$$

$$(b) \tan\left(-\frac{15\pi}{14}\right)$$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

Graphs of Sine and Cosine functions

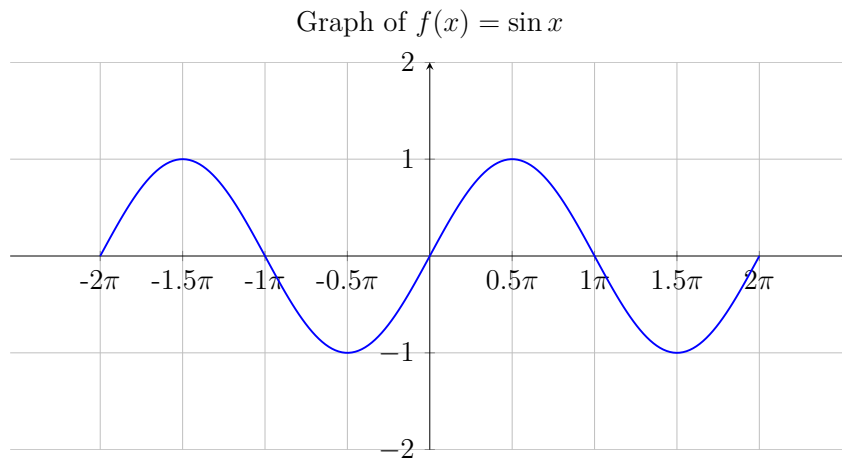
SINE CURVE

The graph of the sine function $f(x) = \sin x$ is called a **sine curve (or sine wave)**. Note that

- Domain of $f = (-\infty, \infty)$.
- f is 2π -periodic.
- Range of $f = [-1, 1]$.
- The sine curve is **symmetric with respect to the origin** because sine function is an odd function.
- x-intercepts: $(n\pi, 0)$ for each integer n
- y-intercept: $(0, 0)$
- The values of f at special and quadrant angles are given as follows

x	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin x$	0	1	0	-1	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0

With the above information the sine curve in the interval $[-2\pi, 2\pi]$ is given below.



COSINE CURVE: The graph of the cosine function $g(x) = \cos x$ is called a **cosine curve(or sine wave)**. We recall that

1. Domain of $g = (-\infty, \infty)$.
2. g is 2π -periodic.
3. Range of $g = [-1, 1]$.
4. The cosine curve is **symmetric with respect to the y-axis** because cosine function is an even function.
5. x-intercepts: $(n\pi + \frac{\pi}{2}, 0)$ for each integer n
6. y-intercept: $(0, 1)$
7. The values of g at special and quadrant angles are given as follows

x	-2π	$-\frac{3\pi}{2}$	$-\pi$	$-\frac{\pi}{2}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\cos x$	1	0	-1	0	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1

With the above information the cosine curve in the interval $[-2\pi, 2\pi]$ is given below.

